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Ontology-Driven Decision Support for Machine Learning Method Selection in Product Development and Production

Combining Semantic Literature Retrieval with Ontology-Based Filtering and Implementation Support

Master's thesis in Engineering and ICT
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ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **NTNU** Norwegian University of Science and Technology

CHAPTER

ONE

INTRODUCTION

This chapter introduces the purpose and overall direction of the thesis. It presents the motivation for the study and describes the main problem that is addressed. The research questions are then defined to guide the work. Finally, the chapter provides an overview of the structure of the thesis.

1.1 Motivation

Machine learning has seen strong growth in recent years and is now widely applied in product development and production. It is used for tasks such as optimization, control, prediction, and decision support [1].

Advances in data availability and computing power have further increased the use of machine learning in industrial settings [2]. As a result, machine learning plays an important role in improving efficiency, supporting decision-making, and enabling new capabilities in engineering and manufacturing.

At the same time, the field of machine learning is large and diverse. Many different algorithms, methods, and approaches are available, each with its own strengths and weaknesses [1]. There is no single method that performs best for all problems, and performance depends on the specific application and data [3, 4].

This creates a challenge for engineers and practitioners, who must select suitable methods for their problems. The large number of available options can act as a barrier and limit the effective use of machine learning in practice [3]. Selecting an appropriate method often requires balancing multiple factors such as accuracy, interpretability, and data requirements [5, 6].

If the selection is not done carefully, the resulting model may perform poorly or fail to provide useful insights [6]. In practice, simpler models are often preferred due to their interpretability, even if more complex models may offer higher accuracy [7]. This highlights the importance of considering trade-offs in real-world applications.

Another challenge is the gap between research and practical implementation. Research results are distributed across many articles and domains, which makes it difficult to gain a complete overview [3]. In addition, most studies focus on developing models, while less attention is given to how they should be selected and applied in practice.

Companies also face challenges related to lack of knowledge and experience when adopting machine learning [8]. In many cases, the selection of methods is based on trial and error, which is time-consuming and inefficient [8]. There is also often pressure to adopt machine learning without a clear plan, which increases the risk of unsuccessful implementations [5].

These challenges highlight the need for more structured approaches that can support method selection and make machine learning more accessible.

1.2 Problem Description

Despite the increasing use of machine learning in manufacturing and engineering, the selection of appropriate methods for specific problems remains insufficiently supported in practice. Existing approaches provide limited guidance on how to match problem characteristics, data properties, and practical requirements with suitable machine learning techniques [9, 10].

A key limitation is the lack of structured decision-support methods for algorithm selection. Current practices often rely on general guidelines or empirical testing, requiring practitioners to evaluate multiple candidate models without clear criteria for narrowing the search space [1, 6]. This makes the process time-consuming and dependent on prior experience, which many users do not have [8].

In addition, existing research primarily focuses on the development and evaluation of machine learning models, while less attention is given to how these methods should be selected and applied in specific industrial contexts [11]. As a result, practitioners lack practical support for early-stage decision-making when defining and approaching machine learning problems.

Another challenge is that relevant knowledge is distributed across a large and fragmented body of literature. Although extensive information exists on machine learning methods and applications, it is difficult to access and use this knowledge in a systematic way during the method selection process [3]. This limits the ability to make informed decisions based on prior research.

Furthermore, there is a lack of tools that connect problem descriptions with machine learning methods in a transparent and interpretable manner. Many existing approaches require detailed technical knowledge of algorithms or rely on simplified selection criteria, which reduces their usefulness for non-experts [10, 8].

These challenges highlight the need for more systematic approaches to method selection. In particular, there is a need for solutions that can connect problem descriptions, machine learning methods, and supporting research.

The large and diverse body of machine learning literature contains valuable knowledge, but it is difficult to use in practice. Combining structured knowledge with information from this literature can support more informed method selection and early-stage implementation.

There is therefore a need for tools that can:

- support method selection based on problem descriptions
- provide interpretable and transparent recommendations
- link methods to relevant research literature
- support early-stage implementation

1.3 Research Questions

Based on the challenges described above, this thesis addresses the following research questions:

- How can ontology-based knowledge improve machine learning method selection?
- How can semantic similarity between problem descriptions and literature support recommendations?
- How can structured knowledge and unstructured text be combined in a single system?
- To what extent can such a system support early-stage machine learning implementation?

To investigate these questions, a web-based decision-support system is developed that combines structured knowledge with information from scientific literature.

1.4 Thesis Structure

The remainder of this thesis is structured as follows:

- Chapter 1 introduces the motivation, problem description, and research questions
- Chapter 2 presents background and related work
- Chapter 3 describes the methodology
- Chapter 4 presents the system implementation
- Chapter 5 presents the results
- Chapter 6 discusses the findings and limitations
- Chapter 7 concludes the thesis and suggests future work

CHAPTER

TWO

BACKGROUND AND STATE OF THE ART

2.1 Machine Learning Method Selection

- Overview of ML paradigms (supervised, unsupervised, etc.)
- Typical ML tasks (classification, regression, time series, etc.)
- Challenges in selecting appropriate methods
- Existing guidelines or frameworks for method selection

2.2 Decision Support Systems

- Systems that support decision-making, not replace it
- Knowledge-based vs data-driven systems
- Often interactive and require user input
- Relevance for engineering and industrial contexts

2.3 Implementation Support for Machine Learning

- Selecting a method is only the first step in ML projects
- Users often need support to start implementation
- Templates and reusable workflows can reduce effort
- Such support can lower the barrier for non-experts

2.4 Ontologies and Knowledge Representation

- What an ontology is (concepts, relations, structure)
- Use of ontologies in engineering and ML domains
- Benefits: structure, explainability, reasoning
- Limitations: manual effort, rigidity

2.5 Semantic Retrieval and Text Similarity

- Text can be represented as embeddings (vectors)
- Similarity measures can compare meaning between texts
- Used to match problem descriptions with relevant content
- More flexible than keyword-based search

2.6 Literature-Based Discovery / Retrieval

- Scientific articles are a rich but unstructured knowledge source
- Challenges include large volume, noise, and unclear relevance
- Methods can be linked to articles through extraction and metadata
- Semantic retrieval can improve matching between problems and literature

2.7 Human-in-the-Loop and Feedback

- User input helps guide and refine recommendations
- Feedback can improve system performance over time
- Requires large user base and careful design to avoid bias

2.8 Web-Based ML Tools / Interfaces

- User interfaces allow interaction with the system
- Should support clear input and understandable output
- Transparency helps users trust recommendations
- Focus is on usability, not detailed UX design

2.9 ML in Product Development and Production

- Go through findings from the literature review in pre-master project

- ML is used for tasks such as Quality prediction and defect detection, Process optimization, Predictive maintenance, Time series forecasting
- Product development often has limited and uncertain data
- Production systems may have structured and continuous data
- Selecting suitable ML methods is challenging in these contexts

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APPENDICES